FANUEL GEBRU

Aberdeen, UK | Github | Email: FisehaFanuel@gmail.com | LinkedIn | Portfolio: https://my-website-fanuel2.vercel.app

EDUCATION

University of Aberdeen | Aberdeen, UK

BSc (Hons) Computing Science, final year, expected graduation July 2027 | On track for an upper 2:1

- **Courses:** Machine Learning and Data Mining, AI, Natural Language Processing, Security, Distributed Systems, Databases, Software Engineering
- **Roles:** Class Representative for international students, Computing Society, AUSA Member

SKILLS

- **Languages:** Python, Java, TypeScript, SQL, C
- **AI and ML:** agentic AI, multi-agent systems, Gemini API, LightGBM, XGBoost, scikit-learn, MLflow
- **Backend:** REST APIs, microservices, event-driven architecture, Spring Boot, FastAPI, NestJS, Flask
- **Data:** Kafka, PostgreSQL, TimescaleDB, Redis, MySQL
- **DevOps:** Docker, GitHub Actions CI/CD, Testcontainers, Linux, Git

EXPERIENCE

VintageET | Co-founder and Backend Engineer | Startup, Addis Ababa (remote) **2025 to present**

- Backend lead on a three-engineer team building a two-sided marketplace, with a NestJS, Prisma and PostgreSQL backend serving a Next.js and TypeScript frontend
- Designed and built REST APIs for listings, orders, search, reviews, chat and notifications, with orders in atomic database transactions so two buyers can't purchase the same item
- Implemented security controls: JWT auth with Google OAuth, role-based access checked against live user state, and presigned uploads to Cloudflare R2
- Set up CI gating lint and type checks, PR code review, and wrote the unit and e2e test suites

Organiser and Workshop Lead | GDG on Campus University of Aberdeen **2026 to present**

- Design and deliver workshops across the UK on building and scaling multiagent systems and multimodal web applications with the Google Gemini API, including "Building Multimodal Web Apps with Gemini API"
- Plan and run hackathons and tech events for students and local developers

Undergraduate Course Demonstrator | University of Aberdeen, School of NCS **2026 to present**

- Ran lab practical sessions for CS1029 Modelling and Problem Solving
- Supported students in breaking problems down and working through them step by step, adapting explanations to different levels

PROJECTS

GridPulse | Real-Time UK Grid Intelligence Platform ***Github.com/cheazer/gridpulse***

Java 21, Spring Boot, Kafka, TimescaleDB, Python, LightGBM, FastAPI, MLflow, Docker

- Built an event-driven pipeline streaming live UK electricity data (National Grid ESO, Elexon, weather) through Kafka into TimescaleDB, with schema-validated events, idempotent writes, and resumable backfills, loading 3.3M records across two years
- Developed day-ahead demand and carbon intensity forecasts with LightGBM and calibrated 80% prediction intervals (conformal prediction), halving error against a seasonal baseline in a 12-month leakage-safe backtest benchmarked against the grid operator's own forecasts
- Shipped a production-style ML service: scheduled forecast job, FastAPI REST API, and an MLflow model registry with a champion/challenger gate that only promotes models that are no worse on unseen data

Staff Workload Management System | UoA Professional Development Project

Python, Flask, PostgreSQL, HTML, CSS, phpMyAdmin, Agile/Scrum

- Co-developed a staff workload management system with a team of 4 developers for the NCS School of Computing. Followed agile software development to build a web app with Python and Flask for the backend and HTML & CSS for the frontend, with a SQL Database managed in phpMyAdmin.
- As the scrum master/project manager, I worked on the database architecture and design, as well as the Flask app for the backend. Guided the team by setting up sprint plans and organising regular meetings with the client.

ACHIEVEMENTS AND ADDITIONAL INFORMATION

- **Awards:** IUPUI Tech Silver Computer Science Award, Indiana University 2024
- **Communication and leadership:** Member of Toastmasters International; Rotaract Club board events organiser (2024); class rep working with AUSA on international student support and inclusion (2024)
- **Languages:** English (fluent), Amharic, Arabic, Tigrinya
- **Interests:** Inclusion and allyship, active interest in international student inclusion at university and in industry